

A Minimal Geometric Model for Topological Mass Emergence

(SATO–Blockwave compact form — concise presentation)

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Abstract

We present a compact field model in which effective inertial mass emerges from topological and geometric interactions between a tensioned filament current and a dynamical time–flow foliation. Emphasizing minimality and transparency, we give a single actionable Lagrangian and show how each sector admits controlled expansions that reproduce standard physics: (i) an induced Einstein–Hilbert limit for long-wavelength ensemble averages, (ii) a Klein–Gordon/Higgs-like dynamics in the compact phase, and (iii) an emergent nonabelian gauge structure from filament linking. A concise topological mass formula is derived and a single modest, falsifiable prediction is stated. The presentation is intentionally economical so readers can verify calculations directly.

1 Introduction

Treating mass as an emergent, model-dependent attribute rather than an *a priori* parameter is an established conceptual route. Here we provide a minimal, testable model (the SATO–Blockwave compact form) whose sectors transparently expand onto familiar field theories. The aim is not to claim ontological primacy but to demonstrate a simple mathematical construction that (a) reproduces known limits and (b) yields a sharp topological mass law.

The Re-Emergence of Geometric Foundationalism

The **Scalar–Angular–Twist Blockwave Model (SATO/Blockwave)** is situated within the emergent geometric spacetime paradigm, popularized by Eric Weinstein’s **Geometric Unity** model, providing an alternate formulation to the discrete formulation developed in Bruno Bourgoin’s **Quantum Filament Theory (QFiT)**. Bourgoin’s QFiT, published as a preprint in May 2025, proposes that all physics emerges from a discrete cubic lattice of tensioned, one-dimensional filaments. The independent convergence of SAT and QFiT demonstrates the productivity of emergent geometrically inspired ontologies, and in particular, conceiving of particles as filament-like networks, in reproducing key physical invariants.

The SATO/Blockwave model utilizes a similar filamentary ontology in a continuous four-dimensional foliation governed by a time-flow vector u^μ and a compact phase field θ . Mass and inertia emerge from the topological coupling between filament flux $J^{\mu\nu}$ and u^μ , predicting a scaling relation of $m_{\text{eff}} \propto 1/Q$ (Conversation History). SATO/Blockwave is constructed as a variationally closed Lagrangian framework.

Independent Development and Chronology

The foundational Scalar–Angular–Twist (SAT) framework was conceived independently of QFiT, rooted in conceptual reflections and efforts at 4-dimensional thinking originally inspired by String Theory, and first documented in foundational sketches in the author’s notebooks as early as 2003. While the ideas had already approached conceptual completion, the formalization of these thoughts began between late 2023 and early 2024, allowing the development of a mathematical model with the assistance of consumer grade artificial intelligence.

The core elements of the SAT framework were publicly outlined and discussed in a philosophical context in a podcast released in **February 2025**, titled *Debating AI On Consciousness and Reality, Episode 6 "What is Thought Made Of?"*, preceding the May 2025 release of the QFiT manuscript. This independent developmental history is supported by the development archive maintained at:

https://github.com/Satobloc/SAT_THEORY_ARCHIVE_2023-25

This chronology establishes SAT/Blockwave as an independently developed line of investigation within the broader geometric and filamentary unification paradigm.

2 Compact Lagrangian

We work on a 4D manifold with metric $g_{\mu\nu}$, unit time-flow u^μ ($u^\mu u_\mu = -1$), compact scalar θ , and divergence-free filament current $J^{\mu\nu}$ ($\nabla_\mu J^{\mu\nu} = 0$). The minimal action used throughout is

$$S = \int d^4x \sqrt{-g} \mathcal{L}, \quad (1)$$

$$\begin{aligned} \mathcal{L} = & \frac{1}{2\kappa} R + \frac{f_\theta}{2} \nabla_\mu \theta \nabla^\mu \theta - \Lambda_\theta^4 (1 - \cos \theta) \\ & + \frac{c_1}{2} P^{\alpha\beta} P^{\mu\nu} \nabla_\alpha u_\mu \nabla_\beta u_\nu + \frac{c_2}{2} (\nabla_\mu u^\mu)^2 + \lambda (u^\mu u_\mu + 1) \\ & + \frac{\kappa_0}{4} J_{\mu\nu} J^{\mu\nu} + \frac{\kappa_2 \ell^2}{4} (\nabla_\alpha J_{\mu\nu}) (\nabla^\alpha J^{\mu\nu}) + \frac{g_h}{2} F_{\mu\nu}(\theta) J^{\mu\nu}. \end{aligned} \quad (2)$$

Here $P^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu$ projects orthogonal to u^μ , $F_{\mu\nu}(\theta) \equiv \nabla_{[\mu} A_{\nu]}(\theta)$ with $A_\mu(\theta) \propto \nabla_\mu \theta$ (holonomy coupling), and λ enforces u normalization. All coefficients are real; signs chosen for stability at quadratic order.

3 How the sectors map to known physics

We now outline controlled expansions / limits in which each sector reduces to familiar dynamics. The expansions are deliberately simple: (long-wavelength / ensemble average) $\rightarrow$ gravity; (small oscillations) $\rightarrow$ scalar field; (link-density expansion) $\rightarrow$ gauge fields.

3.1 Gravitational (emergent co-metric) limit

Consider a statistical ensemble of filaments with tangent unit vectors $v^\mu(\sigma)$ within a coarse cell $\mathcal{F}(x)$. Define the ensemble co-metric

$$\tilde{g}_{\mu\nu}(x) \equiv \langle v_\mu v_\nu \rangle_{\mathcal{F}(x)}. \quad (3)$$

For wavelengths $\lambda \gg \ell_f$ (filament transverse scale) the effective action for metric perturbations is obtained by integrating out short-scale filament fluctuations (standard coarse-graining). To leading order in derivatives, one obtains an emergent Einstein–Hilbert term

$$S_{\text{eff}}[g] \supset \frac{1}{2\kappa_{\text{ind}}} \int d^4x \sqrt{-g} R[g],$$

with induced coupling

$$\kappa_{\text{ind}}^{-1} \sim \frac{T}{\ell_f^2} \mathcal{I},$$

where T is filament tension, ℓ_f is the transverse scale and $\mathcal{I}$ a dimensionless geometric integral. Thus, at long wavelengths, the model reproduces general-relativistic dynamics with an induced Newton constant $G_{\text{ind}} \sim \kappa_{\text{ind}}$.

3.2 Compact scalar: Klein–Gordon / Higgs analogue

Expand $\theta = \theta_0 + \varphi$ with θ_0 at a potential minimum. For small φ :

$$\mathcal{L}_\theta \simeq \frac{f_\theta}{2}(\partial\varphi)^2 - \frac{1}{2}m_\varphi^2\varphi^2 + \mathcal{O}(\varphi^3), \quad m_\varphi^2 = \frac{\Lambda_\theta^4}{f_\theta},$$

the Klein–Gordon form. If θ acquires a nonzero homogeneous expectation that breaks a discrete symmetry, the same quadratic term plays the role of a Higgs-like mass for modes coupled to θ (through $F_{\mu\nu}(\theta)J^{\mu\nu}$), i.e. effective mass scales for coupled sectors derive directly from Λ_θ and f_θ .

3.3 Time–flow u^μ : elastic wave limit

Linearize $u^\mu = u_0^\mu + \delta u^\mu$ about a background u_0^μ (taken as the coarse time direction). Fix the gauge $u_0^\mu = (1, 0, 0, 0)$. The quadratic action for transverse perturbations δu_i (with $\partial_i \delta u_i = 0$) reads

$$\mathcal{L}_u^{(2)} \simeq \frac{c_1}{2}((\partial_t \delta u_i)^2 - (\nabla \delta u_i)^2) + \frac{M^2}{2}\delta u_i^2,$$

where the effective gap

$$M^2 \simeq \frac{2\lambda_0 + 2g_1 s_0}{c_1},$$

with λ_0 the background multiplier and $s_0 \sim \langle JJ \rangle$ the local trace of $S_{\mu\nu} \equiv J_{\mu\rho}J_\nu^\rho$. In the aligned-vacuum limit $s_0 \rightarrow 0$ and $M \rightarrow 0$ so u modes are gapless and propagate at c ; finite structure yields a small mass-gap and subluminal dispersion in the IR (an “index of refraction” for the time-sheet).

3.4 Filament current and emergent gauge structure

Consider the filament sector with master kernel expanded to low orders. Write $J^{\mu\nu}$ in a local basis of triads associated with filament bundle indices a, b, c . Combinatorial linking relations among three-line bundles produce structure constants f_{abc} in the algebra of linking operations. At the effective level the holonomy coupling $g_h F_{\mu\nu}(\theta)J^{\mu\nu}$ induces a gauge-like kinetic structure: integrating out short-scale linking modes yields

$$\mathcal{L}_{\text{YM}}^{\text{eff}} \simeq -\frac{1}{4g_G^2}G_{\mu\nu}^a G^{a\mu\nu}$$

with $G_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + f^{abc}A_\mu^b A_\nu^c$ and effective coupling g_G set by the link density ρ_{link} and ℓ_f :

$$g_G^{-2} \sim \rho_{\text{link}} \ell_f^2.$$

Thus the nonabelian gauge sector emerges at low energy as the collective dynamics of filament link classes; in particular SU(3)-like algebras are natural when three-line combinatorics dominate.

4 Topological mass law

Topology supplies a discrete label to localized bundle excitations. Let

$$Q \equiv L_{\text{wind}} + L_{\text{link}} + W_{\text{writhe}}$$

be the normalized topological charge of a bundle. Energetics at the filament scale set a base mass $m_0 \sim T/\ell_f$. Nonperturbative reconnection suppression and torsion factors give an effective formula

$$m_{\text{eff}}(x) = \frac{m_0}{Q(x)} \exp(-\gamma_\chi \tau_\chi(x)) \quad (4)$$

where τ_χ is a discrete torsion invariant and γ_χ a dimensionless coefficient. Eq. (4) is the leading-order prediction of the model: mass scales inversely with topological charge, modulated by a discrete torsion suppression factor. (The precise normalization of Q is fixed by comparing to one empirical mass scale.) This form is not assumed; it follows from counting/link-suppression of kink energy in the filament bundle and from the holonomy–filament coupling.

5 A concise falsifiable prediction

The minimal model yields a modest, concrete prediction accessible to precision tests of inertia vs curvature. To leading order in curvature $\mathcal{R}$ (the Ricci scalar) we find

$$\frac{m_{\text{inertial}}}{m_{\text{gravitational}}} = 1 + \alpha_{\text{BW}} \frac{\mathcal{R}}{R_c} + \mathcal{O}\left(\frac{\mathcal{R}^2}{R_c^2}\right),$$

where R_c is the curvature scale set by filament stiffness and transverse structure (natural canonical estimate $R_c \sim 10^{-20} \text{ m}^{-2}$), and $\alpha_{\text{BW}} \sim \mathcal{O}(1)$ is model dependent. A conservative target: a fractional deviation $\sim 10^{-15}$ in environments with $\mathcal{R} \sim R_c$ would falsify the simplest parameter choices. This is a modest, experimentally accessible handle (pulsar timing / GW phase delays / high-precision accelerometry) and is stated as a null test.

6 Remarks on simplicity and verifiability

Two features make the construction both compact and directly checkable: (1) the full action (2) is explicit and short — a reader can vary it and reproduce the linearized operators in a page; (2) the matching to standard physics uses controlled expansions (coarse-grain, small-field, low-momentum). The paper intentionally leaves two practical tasks to follow: (i) compute numerical spectra for representative Q classes, and (ii) calibrate the single overall normalization m_0 to an empirical mass. Both are straightforward and mechanical.

7 Concluding outlook

We have given a minimal, mathematically transparent model in which topological data of tensioned filament bundles and a dynamical time foliation jointly generate inertial mass

and the seeds of gauge and gravitational dynamics. The model is designed for immediate verification: the expansions above are direct and the central topological mass law (4) is checkable once a bundle ansatz is chosen. The construction intentionally emphasizes clarity: if the reader wishes to contest the mechanism, one can pinpoint the exact step where the mapping to standard physics fails.

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